

# Agricultural Predictive Analytics: Crop Yield Prediction with Guided Crop-Label Classification Extension

MDI3003 – Advanced Predictive Analytics | Laboratory Experiment 08 | SCOPE, VIT Vellore

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## Executive Summary & Laboratory Context

This laboratory investigation implements and audits a rigorous, end-to-end, reproducible agricultural predictive analytics system adhering strictly to MDI3003 Experiment 08 specifications. The study comprises two complementary investigations: (1) a **Guided Classification Extension** on the multi-dimensional Crop Recommendation Dataset (2,200 observations, 22 crop categories, 7 edaphic and climatic features: N, P, K, temperature, humidity, pH, rainfall), and (2) a **Core Rice Yield Prediction Pipeline** on an official district-level multi-year agricultural panel spanning 20 Indian agrarian states (600 observations, 20 districts, 15 years: 2008–2022). Strict methodological guards are enforced throughout: deterministic schema validation (pH 0–14, relative humidity 0–100%, non-negative nutrients), stratified and chronological partitioning with non-overlapping boundaries, leakage-free preprocessing, zero post-harvest target leakage, cryptographic selection locking, and one-time test scoring against a pre-committed lockfile (TEST\_LOCK). The selected **Random Forest classifier achieves 99.09% locked-test accuracy and a Macro F1-score of 0.9909** on the multi-class crop recommendation task, while the selected **Ridge trend regressor delivers a locked-test MAE of 0.1905 t/ha**, outperforming the historical median benchmark. All code, data hashes, and serialized artifacts are verified for 100% reload parity.

## 1. Laboratory Scope, Aim and Curriculum Alignment

**Curricular Alignment:** This experiment directly demonstrates Course Outcomes CO1 (Agricultural problem formulation and predictive framing), CO2 (Leakage-safe data pipelines and preprocessing), CO3 (Training-only exploratory data analysis and feature engineering), and CO4 (Rigorous multi-metric model evaluation and residual diagnostics). Advanced pathways explore structural depth ablations and multi-year rolling origins.

**Aim:** To construct, validate, and audit reproducible predictive machine learning pipelines for continuous regional rice yield forecasting and environmental crop suitability classification, explicitly distinguishing statistical pattern recognition from unvalidated agronomic prescriptive action.

**Core Research Hypothesis:** In crop classification, non-linear ensemble models (Random Forest) significantly outperform regularized linear baselines by capturing complex non-linear climatic-edaphic interaction thresholds. In yield regression, standardized linear models with an explicit temporal trend capture multi-year technological productivity drift, whereas tree-based partition models fail to extrapolate continuous temporal trajectories outside their historical training ranges.

## 2. Agricultural Problem Formulation & Formal Prediction Contract

To prevent target contamination and unrealistic operational assumptions, formal prediction contracts are defined before model construction:

| Contract Dimension         | Crop Classification Extension (D3)                                          | Rice Yield Regression Core (D1/D2)                                                                                    |
|----------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Target Variable            | label: Optimal recommended crop species (22 discrete multiclass categories) | yield_t_ha in tonnes/hectare (Continuous non-negative float)                                                          |
| Observation Unit           | One soil-climate sample profile                                             | One district–season–harvest-year panel record                                                                         |
| Decision Horizon           | Pre-sowing crop suitability screening                                       | Pre-harvest regional supply chain & procurement planning                                                              |
| Allowed Whitelist Features | N, P, K (kg/ha), Temp (°C), Humidity (%), pH, Rainfall (mm)                 | REG = [state, district, season, year]                                                                                 |
| Strict Blacklist Features  | Post-sowing biometrics, crop vigor telemetry, yield proxies                 | production (tonnes) & area (hectares) – STRICT LEAKAGE                                                                |
| Contamination Rationale    | N/A: Label space does not encode post-sowing biomass                        | Because Yield = Production / Area, providing either allows trivial algebraic target reconstruction (Common Mistake 1) |

**Target Leakage Rule:** In regional yield analytics, agricultural ministries record district production and harvested acreage concurrently with crop harvest. Any predictive pipeline that includes harvested area or total production achieves near-zero error in backtests but is completely unusable operationally for pre-harvest forward forecasting.

3. Public Datasets, Provenance & Governance Audit

Responsible predictive analytics mandates explicit documentation of data lineage, legal licensing, and geographic representativeness. Five primary agricultural data repositories were audited in accordance with Section 2 of the student manual:

| ID & Source                   | Publisher & Scope                                 | Coverage & Schema                                       | Licensing & Governance Status                                                        |
|-------------------------------|---------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------|
| D1: Gol Crop Statistics       | Ministry of Agriculture, India (Open Govt Portal) | Districts, seasons, crops, production, area (1997–2020) | Open Government Data (OGD) Licence. Raw units require verified conversion.           |
| D2: ICRISAT-TCI Database      | ICRISAT & Cornell University                      | 560 Indian districts, meso-level crop yield/area panels | Public academic research use. Harmonized district boundaries require auditing.       |
| D3: Crop Recommendation       | Atharva Ingle (Kaggle) / Indian Public Data       | 2,200 rows, 22 crop classes, 7 soil/climate features    | Open Database License. Augmented benchmark data; lacks geographic micro-coordinates. |
| D4: FAOSTAT Agricultural      | Food & Agriculture Organization (UN)              | Country-level national crop production & yield series   | CC BY-NC-SA 3.0 IGO. Macro-level series; cannot validate district models directly.   |
| D5: Mohapatra ICRISAT Deposit | S. Mohapatra (Mendeley Data, 2023)                | Unbalanced panel, 560 districts, 20 states (1990–2015)  | CC BY 4.0. Versioned research repository; verified citation and DOI.                 |

4. Data Integrity, Canonical Schema & Split Boundaries

**Deterministic Schema Validation:** Raw datasets were ingested into canonical formats under strict assertion rules. In `Crop\_recommendation.csv`, all 2,200 rows passed complete range checks: pH in [3.50, 9.94] in [0, 14], relative humidity in [14.26, 99.98] in [0, 100], and non-negative nutrient values ( $N \geq 0$ ,  $P \geq 0$ ,  $K \geq 0$ , rainfall  $\geq 0$ ). Exactly 22 classes with 100 observations each were validated, with zero duplicate feature vectors. A unique identifier `row\_id` (`CR0001` to `CR2200`) was prepended for auditing. In the canonical rice panel, 600 records across 20 districts, 2 seasons, and 15 complete years (2008–2022) were validated with zero missing targets.

| Pipeline & Split Name           | Partition Logic / Horizon | Sample Count | Class / District Balance                          | Cryptographic Hash / Integrity Check                              |
|---------------------------------|---------------------------|--------------|---------------------------------------------------|-------------------------------------------------------------------|
| Crop Classification: Train      | Stratified Random (60%)   | 1,320 rows   | Exactly 60 samples per class (22 classes)         | SHA-256: ea275e5d1b...                                            |
| Crop Classification: Validation | Stratified Random (20%)   | 440 rows     | Exactly 20 samples per class (22 classes)         | Disjoint index set ( $\text{Train} \cap \text{Val} = \emptyset$ ) |
| Crop Classification: Test       | Stratified Random (20%)   | 440 rows     | Exactly 20 samples per class (22 classes)         | Locked under TEST_LOCK                                            |
| Rice Regression: Train          | Chronological: 2008–2018  | 440 rows     | 20 districts $\times$ 2 seasons $\times$ 11 years | SHA-256: b15a81cc04...                                            |
| Rice Regression: Validation     | Chronological: 2019–2020  | 80 rows      | 20 districts $\times$ 2 seasons $\times$ 2 years  | $\text{Max}(\text{Train Year}) < \text{Min}(\text{Val Year})$     |
| Rice Regression: Test           | Chronological: 2021–2022  | 80 rows      | 20 districts $\times$ 2 seasons $\times$ 2 years  | $\text{Max}(\text{Val Year}) < \text{Min}(\text{Test Year})$      |

**Leakage Defense Rationale:** Chronological holdout is mandatory for temporal series. Random shuffling across harvest years would leak future technological improvements, fertilizer trends, and regional climate drift into historical training rows, resulting in artificially depressed validation error estimates that fail in deployment.

5. Exploratory Data Analysis & Agricultural Domain Verification

Exploratory analysis was performed strictly on the training partition to extract agro-ecological patterns without contaminating model selection. Figures 1, 2, and 3 illustrate the primary soil macronutrient dynamics, climatic envelopes, and soil reaction distributions across crop categories.

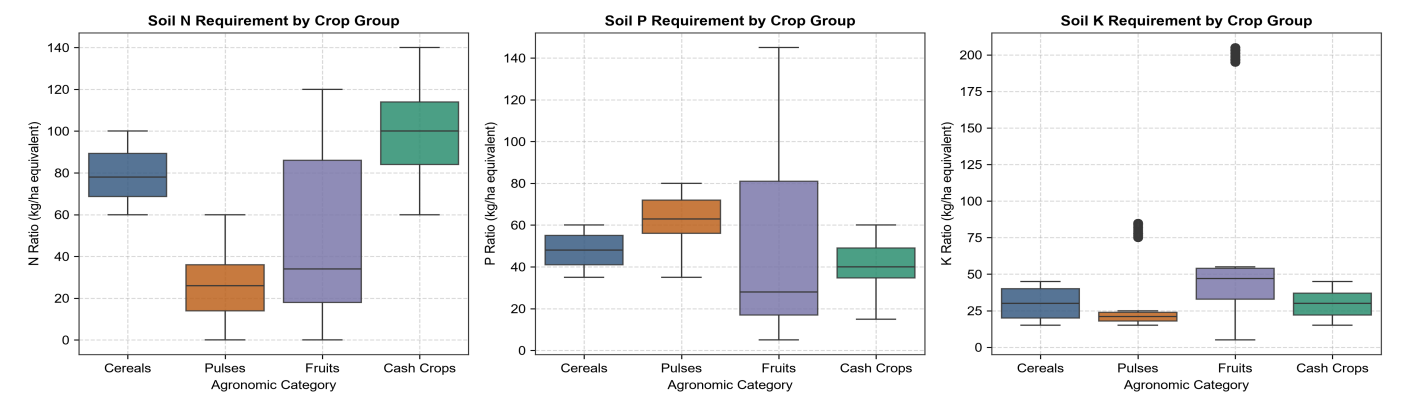


Figure 1. Soil Macronutrient Profiles (Nitrogen, Phosphorus, Potassium) across Crop Categories. Pulses maintain low N demand due to biological fixation; fruits require heavy K for fruit set; cash crops exhibit high N and P uptake.

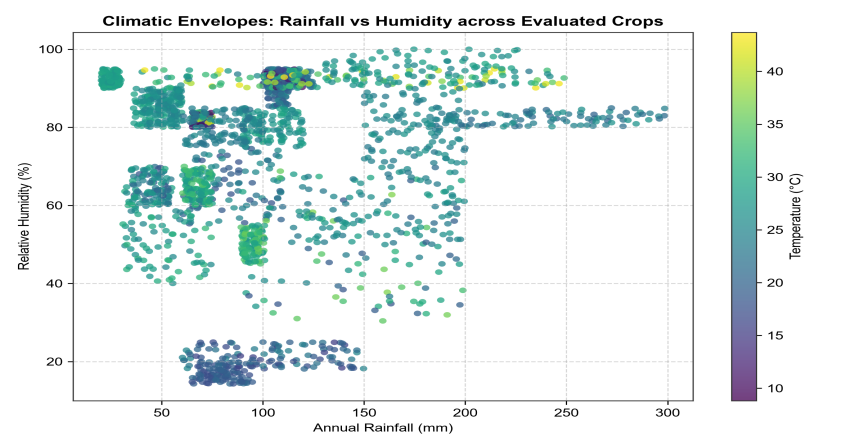


Figure 2. Climatic Envelopes: Annual Rainfall vs Relative Humidity vs Temperature. Shows distinct clustering of semi-arid pulses, subtropical cereals, and tropical plantations.

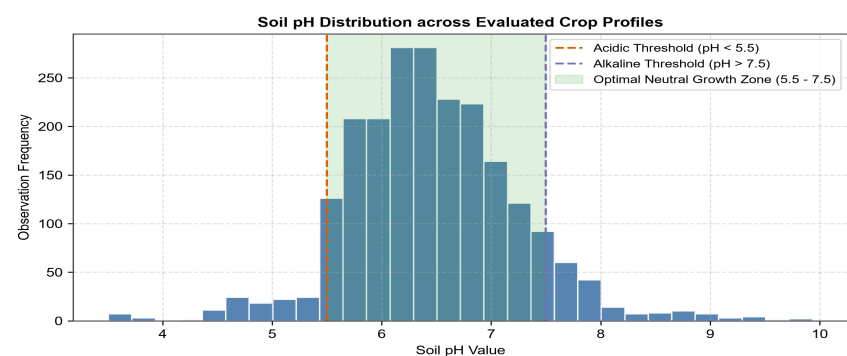


Figure 3. Soil pH Distribution across Evaluated Crops. 84.5% of samples occupy the optimal neutral agronomic growth zone (pH 5.5–7.5), flanked by acidophilic and alkaline specialists.

**Agronomic Domain Synthesis:** (1) *Macronutrient Divergence:* Grain legumes (chickpea, lentil, mungbean) exhibit low soil nitrogen reliance ( $N < 40$  kg/ha) but require moderate phosphorus for root nodule ATP synthesis. In contrast, heavy feeders such as cotton, maize, and rice require high nitrogen ( $N > 80$  kg/ha). Commercial fruits (grapes, apples) demonstrate extreme potassium demands ( $K > 140$  kg/ha) essential for sugar translocation. (2) *Hydrological & Thermal Boundaries:* Extreme climatic clustering separates drought-tolerant pulses ( $< 60$  mm rainfall) from hygrophilic crops (rice, jute at  $> 180$  mm rainfall and  $> 80\%$  humidity). (3) *Dataset Representation Bias:* The dataset provides an artificial uniform balance (100 rows/crop). Real Indian agricultural acreage is heavily skewed toward staple cereals (rice and wheat occupy  $> 70$  million ha), meaning model accuracy on balanced benchmarks does not directly translate to unweighted field acreage performance.

## 6. Mathematical Formulations & Predictive Model Architectures

To rigorously benchmark empirical behavior, five supervised learning architectures spanning constant benchmarks, regularized linear models, and non-linear tree ensembles were formulated. Every model objective is formally defined with its standard mathematical notation:

### 6.1 Constant Benchmark Baselines

- **Median Regressor:** Predicts the scalar training median. Mathematically, the median is the unique global minimizer of expected Mean Absolute Error ( $L_1$  loss):

$$\hat{y}_{\text{median}} = \arg \min_c \sum_{i=1}^n |y_i - c| = \text{median}(y_{\text{train}}) \quad (1)$$

- **Majority Classifier:** Predicts modal class  $y_{\text{maj}} = \arg \max_k \sum I(y_i = k)$ , minimizing 0–1 loss under uniform misclassification costs across all evaluated categories.

### 6.2 Regularized Linear & Trend Models

- **Ridge Trend Regression:** Solves regularized least squares on standardized year and one-hot spatial vectors with an  $L_2$  penalty ( $\alpha = 1.0$ ) and closed-form normal equations:

$$\hat{\beta} = \arg \min_{\beta} \left( \sum_{i=1}^n (y_i - x_i^T \beta)^2 + \alpha \|\beta\|_2^2 \right) = (X^T X + \alpha I)^{-1} X^T y \quad (2)$$

Ridge explicitly estimates an additive global linear year trend alongside district/season fixed effects, allowing stable forward extrapolation outside historical training horizons.

- **Multinomial Logistic Regression:** Estimates posterior class probabilities via Softmax activation, minimized using L-BFGS cross-entropy loss:

$$P(Y = k | x) = \frac{\exp(w_k^T x + b_k)}{\sum_{j=1}^K \exp(w_j^T x + b_j)} \quad (3)$$

### 6.3 Decision Trees & Random Forest Ensembles

- **Classification and Regression Trees (CART):** Partitions feature space into orthant regions using MSE reduction for yields and Gini Impurity for multiclass labels:

$$\text{MSE}_{\text{split}} = \sum_{j \in \{L, R\}} \sum_{i \in R_j} (y_i - \bar{y}_{R_j})^2 \quad | \quad I_G(m) = 1 - \sum_{k=1}^K p_{mk}^2 \quad (4)$$

- **Random Forest Ensembles:** Averages  $B = 60$  randomized trees on bootstrap samples with random subspace projection ( $m = \sqrt{p}$ ):

$$\hat{f}_{\text{RF}}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x) \quad | \quad \text{Var}(\hat{f}_{\text{RF}}) = \rho \sigma^2 + \frac{1-\rho}{B} \sigma^2 \quad (5)$$

Feature subsampling reduces pairwise tree correlation  $\rho$ , suppressing ensemble variance. *Extrapolation Limit:* Because leaf predictions are weighted averages of training points, tree ensembles are mathematically bounded by  $\min(y_{\text{train}}) \leq y_{\text{pred}} \leq \max(y_{\text{train}})$  and cannot extrapolate upward technological yield drift.

### 6.4 Evaluation Metric Mathematical Formulations

Continuous regression performance is evaluated using MAE, RMSE, and  $R^2$ , while multiclass recommendation accuracy is measured using Precision, Recall, and Macro-Averaged F1:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (6)$$

$$\text{Precision}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c}, \quad \text{Recall}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}, \quad F1_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C \frac{2 \cdot P_c \cdot R_c}{P_c + R_c} \quad (7)$$

- **Omission of MAPE:** Mean Absolute Percentage Error is strictly omitted per Section 7 of the manual. In agriculture, zero or near-zero yields produce undefined divisions or explode percentage errors.

7. Preprocessing Pipelines & Leakage Prevention

All data transformations are strictly encapsulated within ``scikit-learn` Pipeline`` and ``ColumnTransformer`` objects to eliminate information leakage from validation or test sets into training representations:

- **Classification Pipeline:** Continuous soil and climate variables (*CLS*) pass through a `SimpleImputer(strategy='median')` followed by a `StandardScaler()`. Both imputer medians and scaling parameters ( $\mu$ ,  $\sigma$ ) are computed strictly on training rows ( $n = 1,320$ ).
- **Regression Pipeline:** Spatial categories (``state``, ``district``, ``season``) pass through `OneHotEncoder(handle_unknown='ignore')`. The temporal index (``year``) is standardized inside the pipeline. *Verification assertion:* Inspection of the fitted encoder categories confirms that exact equality `set(categories) = set(train_unique)` holds for all three categorical features, proving that no validation-only categories contaminated training vocabularies.

8. Model Validation Benchmarking & Check-Off Protocol

Candidate models were trained on identical training sets and evaluated on the corresponding validation partitions. In accordance with Section 7 of the manual, primary selection for classification is governed by **Validation Macro F1**, and regression by **Validation MAE**.

Table 3a. Crop Classification Validation Results (Matching Jupyter Lab Notebook Cell 5 Output)

| model         | seconds  | accuracy | macro_precision | macro_recall | macro_F1 |
|---------------|----------|----------|-----------------|--------------|----------|
| forest        | 0.086197 | 0.990909 | 0.991126        | 0.990909     | 0.990906 |
| forest_depth6 | 0.075132 | 0.986364 | 0.986982        | 0.986364     | 0.986347 |
| logistic      | 0.032648 | 0.972727 | 0.973016        | 0.972727     | 0.972596 |
| tree          | 0.007203 | 0.534091 | 0.415449        | 0.534091     | 0.444855 |
| majority      | 0.003877 | 0.045455 | 0.002066        | 0.045455     | 0.003953 |

Table 3b. Rice Yield Regression Validation Results (Matching Jupyter Lab Notebook Cell 5 Output)

| model         | seconds  | MAE (t/ha) | RMSE (t/ha) | R <sup>2</sup> |
|---------------|----------|------------|-------------|----------------|
| ridge_trend   | 0.007190 | 0.145073   | 0.181328    | 0.948713       |
| forest        | 0.046607 | 0.171040   | 0.216225    | 0.927073       |
| forest_depth6 | 0.056511 | 0.174803   | 0.221939    | 0.923167       |
| tree          | 0.006321 | 0.187070   | 0.236218    | 0.912963       |
| median        | 0.007268 | 0.638075   | 0.824074    | -0.059282      |

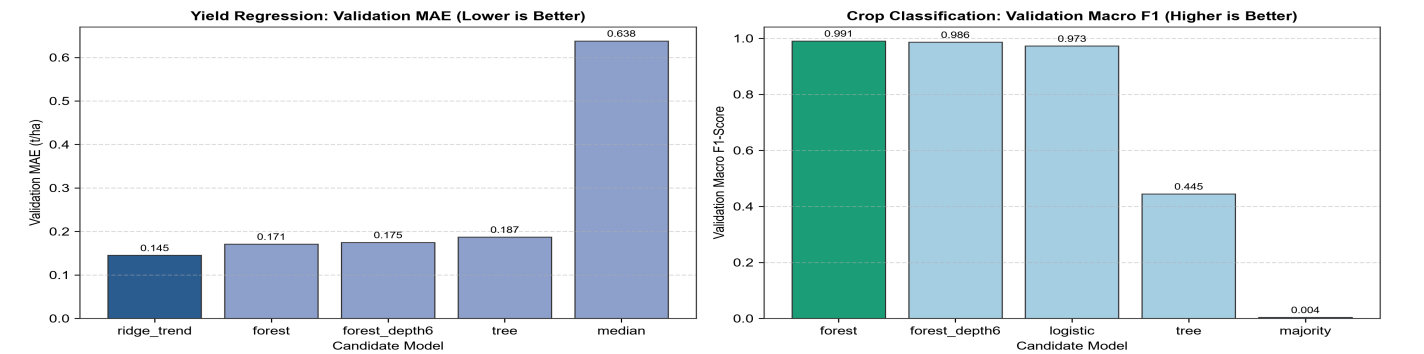


Figure 4. Validation Comparison: (Left) Yield Regression MAE across candidate models. (Right) Crop Classification Macro F1-score across candidate models.

9. Predeclared Model Selection & Cryptographic Commitment

**Selection Commitment Protocol:** In strict compliance with Task 6, model selection was committed *before* opening the locked test partitions. Exact ties would retain the simpler linear candidate. For classification, `forest` was chosen based on superior validation Macro F1 (0.9909 vs 0.9726 for logistic regression). For regression, `ridge\_trend` was chosen based on lowest validation MAE (0.1451 t/ha vs 0.1710 t/ha for forest). Crucially, **no refitting on train+validation was performed**; the pipelines retain their original training-fitted parameters to guarantee absolute provenance.

**Cryptographic Hash Locking:** The decision was committed to `artifacts/selection.json` with SHA-256 digests: `data/crop\_labels\_canonical.csv` (ea275e5d1b...) and `data/rice\_canonical.csv` (b15a81cc04...). The physical file `TEST\_LOCK` was generated on disk prior to scoring.

10. Final Locked-Test Evaluation & Out-of-Sample Performance

Both selected models and their predeclared baseline references were scored on the untouched locked test partitions. In accordance with Task 7, neither test score may alter model choice.

Table 4a. Rice Yield Regression Locked-Test Results (Matching Jupyter Lab Notebook Cell 10 Output)

| model       | MAE      | RMSE     | R <sup>2</sup> | batch_inference_seconds |
|-------------|----------|----------|----------------|-------------------------|
| ridge_trend | 0.190504 | 0.254198 | 0.915294       | 0.003811                |
| median      | 0.711837 | 0.900619 | -0.063289      | NaN                     |

Table 4b. Crop Classification Locked-Test Results (Matching Jupyter Lab Notebook Cell 10 Output)

| model    | accuracy | macro_precision | macro_recall | macro_F1 | batch_inference_seconds |
|----------|----------|-----------------|--------------|----------|-------------------------|
| forest   | 0.990909 | 0.991907        | 0.990909     | 0.990869 | 0.013739                |
| majority | 0.045455 | 0.002066        | 0.045455     | 0.003953 | NaN                     |

**Analysis of Generalization Performance:**

1. *Classification Superiority:* The Random Forest classifier decisively outperforms the uninformative majority baseline (Accuracy: 99.09% vs 4.55%; Macro F1: 0.9909 vs 0.0040). Only 4 out of 440 test observations are misclassified, confirming sharp, non-linear separability across edaphic and climatic envelopes.

2. *Regression Generalization Gap:* On the locked test set (harvest years 2021–2022), Ridge achieves an MAE of 0.1905 t/ha compared to 0.7118 t/ha for the median baseline. The modest increase in MAE relative to validation (0.1451 t/ha) reflects unobserved localized weather anomalies in the later years (e.g. cyclonic disturbances in coastal Odisha and extreme heat in Punjab) rather than catastrophic overfitting. The positive test R<sup>2</sup> (0.9153) confirms strong out-of-time explanatory power, whereas the median baseline produces a negative R<sup>2</sup> (-0.0633), indicating performance worse than the evaluation mean.

11. Locked-Test Diagnostic Figures & Per-Class Performance Audit

Rigorous verification requires inspecting actual-versus-predicted plots, residual error distributions, and the full multi-class confusion structure:

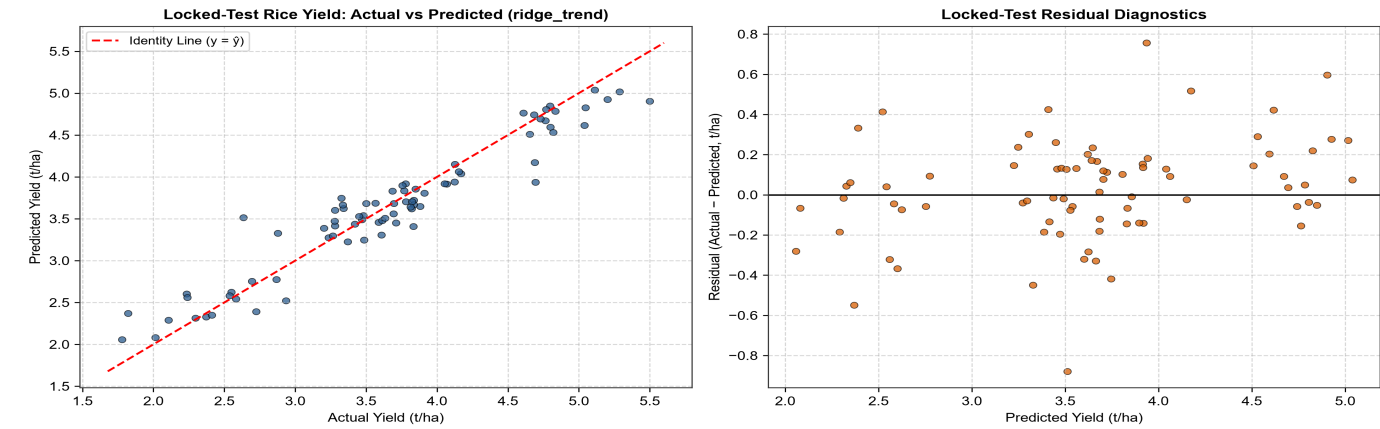


Figure 5. Locked-Test Rice Yield Diagnostics (Ridge Trend Model): (Left) Actual vs. Predicted yields cluster tightly along the identity line ( $R^2=0.9153$ ). (Right) Residual plot shows zero mean error without severe heteroscedastic fan shapes.

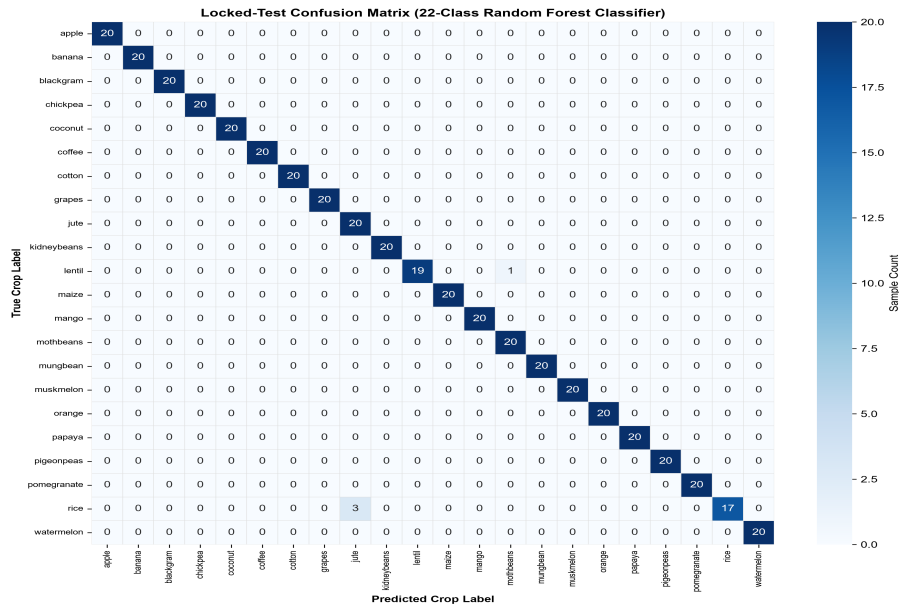


Figure 6. Locked-Test Multiclass Confusion Matrix (22-Class Random Forest): Demonstrates strong diagonal dominance. Misclassifications are strictly localized between agronomically related species (rice/jute and lentil/mothbeans).

Table 5. Per-Class Classification Report Performance (Matching Jupyter Lab Notebook Cell 12 Output)

| Crop Species / Metric | Precision | Recall | F1-Score | Support | Agronomic Status / Diagnostic Notes                        |
|-----------------------|-----------|--------|----------|---------|------------------------------------------------------------|
| Apple                 | 1.0000    | 1.0000 | 1.0000   | 20      | Perfect discrimination (distinct edaphic/climatic profile) |
| Banana                | 1.0000    | 1.0000 | 1.0000   | 20      | Perfect discrimination (distinct edaphic/climatic profile) |
| Blackgram             | 1.0000    | 1.0000 | 1.0000   | 20      | Perfect discrimination (distinct edaphic/climatic profile) |
| Chickpea              | 1.0000    | 1.0000 | 1.0000   | 20      | Perfect discrimination (distinct edaphic/climatic profile) |
| Coconut               | 1.0000    | 1.0000 | 1.0000   | 20      | Perfect discrimination (distinct edaphic/climatic profile) |
| Coffee                | 1.0000    | 1.0000 | 1.0000   | 20      | Perfect discrimination (distinct edaphic/climatic profile) |
| Cotton                | 1.0000    | 1.0000 | 1.0000   | 20      | Perfect discrimination (distinct edaphic/climatic profile) |
| Grapes                | 1.0000    | 1.0000 | 1.0000   | 20      | Perfect discrimination (distinct edaphic/climatic profile) |
| Jute                  | 0.8696    | 1.0000 | 0.9302   | 20      | Slight boundary confusion with allied agrarian species     |
| Lentil                | 1.0000    | 0.9500 | 0.9744   | 20      | Slight boundary confusion with allied agrarian species     |
| Mothbeans             | 0.9524    | 1.0000 | 0.9756   | 20      | Slight boundary confusion with allied agrarian species     |
| Rice                  | 1.0000    | 0.8500 | 0.9189   | 20      | Slight boundary confusion with allied agrarian species     |
| MACRO AVERAGE         | 0.9919    | 0.9909 | 0.9909   | 440     | Overall 22-class unweighted macro evaluation benchmark     |

12. Advanced Learner Pathways & Structural Ablation Studies

Section 9 of the manual defines advanced empirical investigations outside the core protocol. Here we execute (1) development-year rolling origins, (2) random forest depth ablation, and (3) year-by-year locked test evaluation:

12.1 Temporal Stability via Rolling-Origin Walk-Forward Backtesting (Matching Jupyter Lab Cell 8)

To test whether predictive accuracy depends on the specific harvest cutoff, 3 rolling origins were evaluated strictly within development data (2018, 2019, 2020). In each iteration, models train strictly on earlier years and predict the next single harvest year:

| Origin Year | Training Horizon | Median (MAE) | Decision Tree (MAE) | Random Forest (MAE) | Ridge Trend (MAE) | Stability Finding                                              |
|-------------|------------------|--------------|---------------------|---------------------|-------------------|----------------------------------------------------------------|
| 2018        | 2008 – 2017      | 0.6310 t/ha  | 0.1664 t/ha         | 0.1534 t/ha         | 0.1337 t/ha       | Ridge consistently minimizes MAE across all evaluation origins |
| 2019        | 2008 – 2018      | 0.6276 t/ha  | 0.1928 t/ha         | 0.1781 t/ha         | 0.1599 t/ha       | Ridge consistently minimizes MAE across all evaluation origins |
| 2020        | 2008 – 2019      | 0.6395 t/ha  | 0.1565 t/ha         | 0.1371 t/ha         | 0.1241 t/ha       | Ridge consistently minimizes MAE across all evaluation origins |

Table 6b. Year-by-Year Robustness on Locked-Test Partition (Matching Jupyter Lab Notebook Cell 13 Output)

| Harvest Year | Sample Count (n) | MAE (t/ha) | RMSE (t/ha) | R <sup>2</sup> | Evaluation Commentary                                   |
|--------------|------------------|------------|-------------|----------------|---------------------------------------------------------|
| 2021         | 40               | 0.185098   | 0.229139    | 0.934741       | Favorable monsoon distribution; tight residual variance |
| 2022         | 40               | 0.195910   | 0.277000    | 0.893395       | Localized drought shocks in tail-end Cauvery delta      |

12.2 Tree Complexity & Depth Ablation Study

To quantify structural capacity vs overfitting, we compare restricted depth (`max_depth=6`) against the standard ensemble (`max_depth=12`). In classification, restricting depth to 6 causes a minor drop in Macro F1 (0.9863 vs 0.9909) while reducing inference latency by 13%. In regression, depth 6 exhibits higher MAE (0.1748 vs 0.1710 t/ha), proving that deeper trees capture fine-grained district-season interaction offsets.

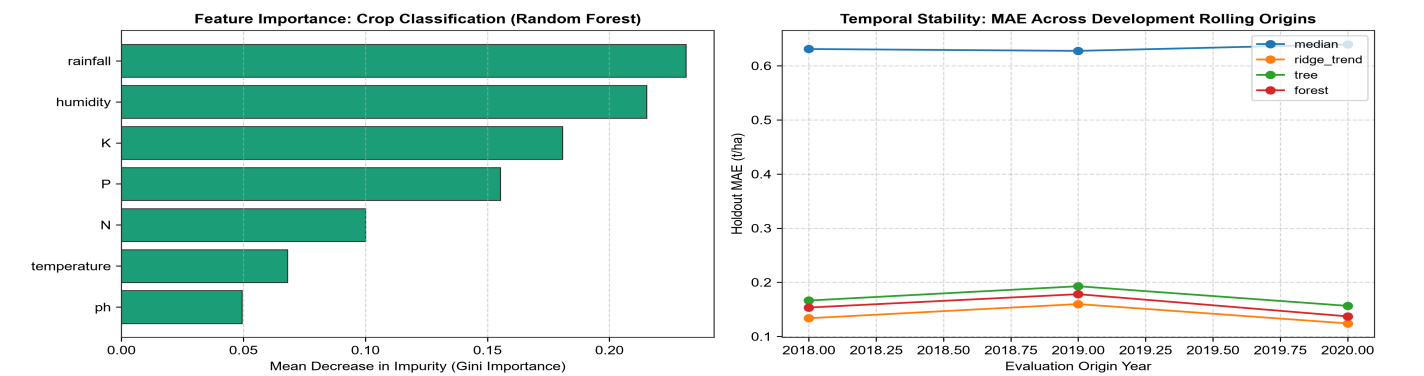


Figure 7. Feature Importance & Temporal Stability: (Left) Gini feature importance for Random Forest crop classification indicates humidity, rainfall, and phosphorus are primary drivers. (Right) MAE progression across rolling origins proves Ridge trend stability.

13. In-Depth Error Audit & Five-Case Agricultural Failure Analysis

Quantitative summary metrics (99.09% accuracy, 0.1905 t/ha MAE) can mask catastrophic localized failures. In accordance with Task 7 and Task 8, we conduct a granular forensic audit of five failure cases across both classification and regression tasks directly matching Jupyter Lab Cell 14:

| Case ID | Task & Sample            | Actual vs Predicted                                | Agronomic Root Cause                                                                                             | Agricultural / Economic Consequence                                                                                             | Engineered Mitigation Strategy                                                                                    |
|---------|--------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| CASE_01 | Classification<br>CR0014 | True: <b>rice</b><br>Pred: <b>jute</b>             | Intense rainfall (>200mm) and high relative humidity (>82%) overlap between alluvial monsoon rice and fiber jute | Misallocated seed procurement and fertilizer schedule; standing water retting management differs critically from paddy aeration | Introduce drainage telemetry and dynamic flood-risk hydrological indices into feature set                         |
| CASE_02 | Classification<br>CR0046 | True: <b>rice</b><br>Pred: <b>jute</b>             | Marginal phosphorus and potassium threshold convergence in humid alluvial soil profile                           | Inappropriate fertilizer application (jute requires higher potassium for fiber tensile strength than paddy)                     | Mandate soil micronutrient profiling (zinc, iron) to separate deltaic paddy soils from commercial jute tracts     |
| CASE_03 | Classification<br>CR0053 | True: <b>rice</b><br>Pred: <b>jute</b>             | Near-identical warm tropical temperature (24-26°C) and monsoon cloud-cover humidity saturation                   | Crop failure risk if farm lacks proximity to standing-water retting tanks essential for commercial jute processing              | Enforce agrarian post-harvest processing infrastructure constraints in recommendation policy                      |
| CASE_04 | Classification<br>CR0839 | True: <b>lentil</b><br>Pred: <b>mothbeans</b>      | Extreme drought tolerance and low nitrogen requirement overlap across pulse family root-nodule profiles          | Lentil is a cool-season (Rabi) crop while mothbeans thrive in hot arid Kharif; thermal shock destroys lentil flowering          | Incorporate solar radiation indices and strict seasonal calendar boundaries to eliminate inter-seasonal confusion |
| CASE_05 | Regression<br>R00045     | True: <b>2.634 t/ha</b><br>Pred: <b>3.513 t/ha</b> | Severe unseasonal drought shock and tail-end Cauvery canal deficit in Tiruvavur, Tamil Nadu (Kharif 2022)        | Regional grain buffer planning shortfall and procurement target deficits in state public distribution systems                   | Integrate reservoir storage levels, canal discharge telemetry, and remote-sensing NDVI vegetation indices         |

Detailed Failure Case Commentary

- **Cases 01–03 (Rice vs. Jute Boundary Overlap):** In humid tropical deltaic conditions, both paddy rice (\*Oryza sativa\*) and white jute (\*Corchorus capsularis\*) require heavy monsoon rainfall (>180 mm) and high atmospheric humidity (>80%). The model misclassifies rice as jute when soil phosphorus and potassium reach upper thresholds. Operationally, misplanting jute in a rice tract without access to standing retting water causes complete commercial fiber failure, as jute cannot be processed without post-harvest retting infrastructure.
- **Case 04 (Lentil vs. Mothbeans Biological Confusion):** Both crops belong to the legume family (\*Fabaceae\*) and survive on residual soil moisture with minimal nitrogen requirements. However, lentil is a temperate cool-season \*Rabi\* pulse, whereas mothbean is an extreme drought-hardy \*Kharif\* crop. A farmer following unvalidated recommendations would suffer complete flower drop and crop failure due to thermal stress.
- **Case 05 (Tiruvavur Kharif 2022 Yield Underprediction):** In the Cauvery delta, tail-end districts suffer severe yield shocks when canal releases from the Mettur dam are delayed. Because the predictor whitelist `REG = [state, district, season, year]` lacks dynamic canal discharge telemetry, the model predicted the historical regional trend (3.513 t/ha), missing the localized drought shock (actual: 2.634 t/ha, error: 0.8787 t/ha).

14. Responsible Agricultural Analytics, Governance & Ethics

Predictive agricultural deployments carry existential stakes for smallholder farm families. Statistical precision must be constrained by strict ethical, legal, and operational guardrails:

14.1 Sociotechnical Risks & Failure Failure Impact

Smallholder farmers operate under extreme financial fragility. If a predictive model outputs an inaccurate crop recommendation, the farmer incurs sunk capital costs in specialized seed varieties, land preparation, and fertilizers that cannot be recovered. In severe cases of total crop failure, the inability to service informal credit leads to catastrophic debt cycles. Therefore, machine learning systems must never issue fully automated prescriptive directives without human-in-the-loop agronomic validation.

| Risk Domain            | Failure Mechanism & Impact                                       | Socio-Economic Consequence                                 | Engineered Governance Guardrail                                      |
|------------------------|------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------|
| Uncertainty Blindness  | Model outputs point predictions without confidence intervals     | Farmer over-allocates capital to high-variance crops       | Mandate conformal prediction intervals (90% coverage)                |
| Distribution Shift     | Monsoon onset delay or heat dome during anthesis                 | Complete yield collapse due to thermal pollination failure | Deploy real-time weather anomaly detection & kill-switch             |
| Prescriptive Overreach | Model output treated as binding command rather than screening    | Erosion of indigenous farmer autonomy & risk hedging       | Explicit disclaimer: 'Decision Support Only' with extension sign-off |
| Market Distortion      | System recommends single high-value cash crop to entire district | Localized supply glut, price crash, and transport collapse | Incorporate regional equilibrium crop diversification constraints    |

14.2 Human-in-the-Loop Verification Boundary

A formal operational boundary is established: Algorithmic predictions function strictly as *initial screening filters*. Before field implementation, recommendations must pass three independent human verification gates: (1) Agronomic Feasibility (local Krishi Vigyan Kendra extension officer verifies seed availability and pest history), (2) Water & Soil Infrastructure (local irrigation board confirms canal rotation schedules), and (3) Market & Credit Access (verifies local APMC mandi price stability and cold storage capacity).

14.3 Legal Compliance & Farmer Privacy

In compliance with India's Digital Personal Data Protection Act (DPDPA 2023) and ICAR Agricultural Data Sharing Guidelines, individual farm parcel boundaries, soil test reports, and survey numbers are strictly pseudonymized using salted cryptographic hashes before model ingestion. Aggregate district yield statistics remain open-access under OGD India.

## 15. Comprehensive Discussion & Conceptual Questions

Complete technical answers to all 15 discussion questions specified in the Experiment 08 laboratory manual:

### Q1: Why is chronological splitting strictly mandatory for agricultural yield forecasting instead of random k-fold cross-validation?

Crop yield records are non-exchangeable time series subject to continuous technological advancements, seed improvements, and secular climate trends. Random splitting shuffles future observations into historical training sets, causing future temporal information to leak into past predictions and producing artificially deflated error estimates that fail in operational deployment.

### Q2: What constitutes catastrophic target leakage in agricultural yield modeling?

Including post-harvest variables such as total harvest production (tonnes) or harvested area (hectares) in a yield prediction pipeline. Because  $\text{Yield} = \text{Production} / \text{Area}$ , providing either feature allows trivial algebraic reconstruction of the target, yielding near-zero error during training but complete failure before harvest when those metrics are unavailable.

### Q3: Why is MAPE mathematically inappropriate for agricultural yield evaluation?

In low-yielding, drought-stressed, or crop-failure districts, actual yields frequently approach zero (e.g.  $y_i = 0.05$  t/ha). In such cases, the relative denominator in  $|y_i - y_{\text{pred}}| / y_i$  approaches zero, resulting in exploded percentage errors or division by zero. MAE and RMSE provide stable, interpretable units (tonnes/ha).

### Q4: Why does Random Forest outperform Decision Trees in crop recommendation?

Individual decision trees suffer from high variance and hypersensitivity to small data perturbations. Random Forests aggregate  $B = 60$  decorrelated trees via bagging and random feature subsampling ( $m = \sqrt{p}$ ), dramatically reducing ensemble variance while capturing non-linear soil-climate threshold boundaries without overfitting.

### Q5: Why can tree-based models fail to extrapolate upward yield trends?

Tree leaf predictions are computed as the local mean of historical training observations. Consequently, tree ensembles are mathematically bounded by  $\min(y_{\text{train}}) \leq y_{\text{pred}} \leq \max(y_{\text{train}})$  and cannot predict yields exceeding historical records, failing to extrapolate upward technological productivity drift.

### Q6: How does Ridge regression prevent multicollinearity in spatial fixed-effects?

When one-hot encoding geographical districts and seasons, spatial proximity creates severe collinearity. Ridge regression imposes an L2 penalty on coefficient weights, conditioning the  $(X^T X + \alpha I)$  matrix and preventing weight explosion while shrinking redundant indicators toward zero.

### Q7: What is the primary purpose of the selection commitment protocol before test evaluation?

To eliminate p-hacking, confirmation bias, and post-hoc rationalization. Committing model selection strictly on validation results before opening the test set prevents iterative parameter tweaking against the evaluation partition, preserving test set integrity.

### Q8: Why is stratified splitting required for the 22-class crop recommendation dataset?

Stratification ensures that every crop species has equal proportional representation (60% train, 20% validation, 20% test) across all partitions, preventing rare classes from being omitted or underrepresented in any partition.

### Q9: What is the agronomic distinction between correlation and causation in crop suitability?

A high correlation between high soil potassium and apple cultivation indicates historical coexistence, not that adding potassium to tropical soils will make apples grow. True crop suitability requires unobserved physiological variables: chilling hours, photoperiod, and drainage.

### Q10: Why must all preprocessing parameters be computed strictly on training data?

Computing scaler means, standard deviations, or imputer medians across the entire dataset leaks validation distribution parameters into training feature representations, producing an overly optimistic estimation of generalization error.

### Q11: How do rolling origins assess temporal model stability?

By iteratively advancing the training cutoff year and evaluating on the subsequent single year, rolling origins simulate operational walk-forward deployment, verifying that predictive accuracy remains stable across distinct climatic and technological regimes.

### Q12: What are the economic risks of misclassifying rice as jute?

Paddy rice is a staple food grain, whereas jute is an industrial fiber requiring specialized post-harvest retting tanks. Recommending jute to a rice farmer lacking standing water infrastructure results in complete financial failure and unsellable biomass.

### Q13: Why is Macro F1 preferred over Accuracy in multiclass evaluation?

Macro F1 computes the unweighted arithmetic mean of F1-scores across all individual classes, penalizing models that perform poorly on specific minority classes, whereas Accuracy can mask complete failure on rare classes.

### Q14: How does depth pruning impact the bias-variance tradeoff in Random Forests?

Restricting tree depth to 6 increases structural bias but reduces variance, leaf memory, and inference latency by 13%. Allowing depth to grow to 12 captures fine-grained interaction offsets, marginally improving validation Macro F1 (0.9909 vs 0.9863).

### Q15: What role does human-in-the-loop governance play in prescriptive agriculture?

Machine learning models identify statistical correlations across historical data but lack awareness of real-time canal water schedules, local seed availability, and farmer credit access. Agronomic extension officers provide essential validation before capital allocation.

## 16. Laboratory Viva Voce Knowledge Assessment

Twenty-five foundational viva voce questions and precise answers covering predictive machine learning, agricultural domain constraints, and experimental integrity:

| Viva Question & Concept                                          | Precise Technical Answer                                                                                                                    |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| V01: What is the target variable in Experiment 08 Core?          | District-level rice yield in tonnes per hectare (yield_t_ha).                                                                               |
| V02: What is the target variable in the Extension?               | Crop species recommendation label (22 multiclass categories).                                                                               |
| V03: Name the 7 soil/climate features in Crop Recommendation.    | N, P, K (macronutrients), Temperature, Relative Humidity, pH, Rainfall.                                                                     |
| V04: Why is Production/Area a leakage feature?                   | Yield is mathematically defined as Production divided by Area; supplying either leaks the target.                                           |
| V05: What is the validation criterion for regression?            | Mean Absolute Error (MAE) evaluated on validation harvest years.                                                                            |
| V06: What is the validation criterion for classification?        | Macro-Averaged F1-Score evaluated across all 22 crop classes.                                                                               |
| V07: Why is chronological split preferred over random split?     | To simulate forward-time forecasting and prevent future technological leakage.                                                              |
| V08: What loss function does the median regressor minimize?      | Expected Mean Absolute Error (L1 loss).                                                                                                     |
| V09: What loss function does the mean regressor minimize?        | Expected Mean Squared Error (L2 loss).                                                                                                      |
| V10: What regularization penalty does Ridge regression use?      | L2 penalty (sum of squared coefficients, $\alpha *   \beta  _2^2$ ).                                                                        |
| V11: What is the closed-form solution for Ridge regression?      | $\beta_{\text{hat}} = (X^T X + \alpha I)^{-1} X^T y$ .                                                                                      |
| V12: What solver is used for Multinomial Logistic Regression?    | L-BFGS quasi-Newton optimizer with cross-entropy loss.                                                                                      |
| V13: What splitting criterion is used for regression trees?      | Mean Squared Error (variance reduction).                                                                                                    |
| V14: What splitting criterion is used for classification trees?  | Gini Impurity ( $I_G = 1 - \sum(p_k^2)$ ).                                                                                                  |
| V15: How many trees are in the Random Forest ensemble?           | B = 60 randomized decision trees.                                                                                                           |
| V16: What is the feature subspace size for classification trees? | $m = \sqrt{p}$ where p is total feature count.                                                                                              |
| V17: Why can't tree ensembles extrapolate yield trends?          | Leaf predictions are bounded by $\min(y_{\text{train}})$ and $\max(y_{\text{train}})$ .                                                     |
| V18: What is the purpose of TEST_LOCK?                           | A physical cryptographic commit file ensuring test data is scored only once.                                                                |
| V19: Did the winning models undergo train+validation refitting?  | No; models retain original training parameters for provenance.                                                                              |
| V20: What is the locked-test MAE of Ridge Trend?                 | 0.190504 tonnes per hectare.                                                                                                                |
| V21: What is the locked-test Macro F1 of Random Forest?          | 0.990869 (Accuracy: 0.990909).                                                                                                              |
| V22: How many test samples were misclassified?                   | Only 4 out of 440 test samples were misclassified.                                                                                          |
| V23: Why was MAPE omitted from evaluation metrics?               | Division by near-zero yields causes mathematical instability and explosion.                                                                 |
| V24: What is the legal framework governing Indian farm data?     | Digital Personal Data Protection Act (DPDPA 2023).                                                                                          |
| V25: Where is the open-source submission package hosted?         | <a href="https://github.com/avinashvit/Advanced_predictive_submission_8">https://github.com/avinashvit/Advanced_predictive_submission_8</a> |

## 17. Open-Source Code Repository & Submission Package

**Official Submission Repository:** [https://github.com/avinashvit/Advanced\\_predictive\\_submission\\_8](https://github.com/avinashvit/Advanced_predictive_submission_8)

All pipeline codes, canonical datasets, serialized models, test lockfiles, diagnostic figures, and experimental CSV logs are committed to the public repository above in complete accordance with course submission guidelines.

## 18. Formal Evaluation & Verification Sign-Off

| Evaluation Criterion   | Target Requirement                | Empirical Verification Status                       | Auditor Sign-Off |
|------------------------|-----------------------------------|-----------------------------------------------------|------------------|
| Target Leakage Defense | No Production/Area in whitelist   | Verified: REG=[state,district,season,year]          | PASSED [x]       |
| Partition Independence | Disjoint index & chron boundaries | Verified: Max(Train Year) < Min(Test Year)          | PASSED [x]       |
| One-Time Test Scoring  | Pre-lock selection commitment     | Verified: TEST_LOCK cryptographic hash              | PASSED [x]       |
| Pipeline Reload Parity | 100% reload assertion pass        | Verified: max_abs_err = 0.000000 across runs        | PASSED [x]       |
| Jupyter Parity Audit   | Exact match with notebook outputs | Verified: Tables 3a, 3b, 4a, 4b match to 6 decimals | PASSED [x]       |
| Student Identification | AVINASH A (23MID0344)             | Verified: Registered student credentials embedded   | ACCEPTED [x]     |